WAYVION

An AI-Powered Personalized & Adaptive Learning Assistant

Technical Report

MSIS 522B - Advanced Business Machine Learning

Swati Choudhury, Sandrine Lepesqueux, Nathan Fitzgerald, Nicholas Ray German

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1. Executive Summary

Learning has never been easier-there are myriad resources on any topic-and that abundance is precisely the problem. Learners face an overwhelming and fragmented landscape: YouTube videos with no clear sequence, Reddit threads with outdated suggestions, and generic roadmap sites that ignore individual background or time constraints.

WayVion solves this by generating fully personalized, adaptive learning roadmaps in seconds. Unlike general-purpose LLMs that produce static, hallucination-prone lists, WayVion operates as a structured full-stack, multi-agent system: it curates real, verified resources from the internet, sequences them pedagogically around the learner's specific goal and constraints, tracks progress, tests understanding through quizzes, and adapts the plan in real time when the learner struggles or falls behind schedule.

WayVion was built end-to-end as the MSIS 549B capstone project and subsequently extended with a Community of Learners layer, a Groups system, and an expanded adaptation engine. It won Runner-Up and Judges' Favorite at the GenAI and Agentic Fair.

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2. Business Problem

The online learning landscape is vast but poorly navigated. A learner who decides to study Machine Learning or Finance faces the same practical question: where do I start, what do I do next, and how do I know I'm on the right path? Existing solutions each fail in a distinct way:

- Search engines: Surface a flood of unordered results with no pedagogical sequence.
- Reddit threads & blogs: Crowd-sourced advice that may be outdated, irrelevant, or biased toward the author's own style.
- Online Learning Platforms (Coursera, Khan Academy, etc.): Static, one-size-fits-all curricula limited to content already curated by the platform.
- General-purpose LLMs (ChatGPT, Claude): Produce static outputs, frequently hallucinate resource links, cannot track progress, and lose context when the conversation ends.

WayVion understands where the learner is in terms of skill proficiency, where they want to go, how much time they have, and their learning goal-and keeps the roadmap alive as reality diverges from the original plan.

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3. What Is WayVion?

WayVion is an AI-powered personalized and adaptive learning assistant. It takes a user's topic, skill level, available time, and learning goal, and generates a fully personalized learning roadmap with curated, real, verified resources sequenced pedagogically.

WayVion acts as a personalized learning navigator, not a content platform. It does not create educational content-it curates the best content that already exists on the internet and sequences it into a coherent, personalized roadmap built from a pedagogical standpoint.

What sets WayVion apart is that roadmaps are not static-they adapt in real time when the learner struggles, falls behind schedule, or changes their learning goals entirely.

How WayVion Differentiates from Other Products

The table below summarizes key differentiators over general-purpose LLMs and generic roadmap tools:

Capability	ChatGPT / Claude	Generic Roadmap Sites	WayVion
Personalized roadmap	Static list	Static list	Dynamic, fully personalized
Real, verified resources	Hallucinated links	Curated but fixed	Live-fetched from web
Adapts to progress	No	No	Yes - quiz + self-report triggered
Respects time constraints	Generic advice	No	Math-based, day-by-day schedule
Tracks learning style	No	No	4 learning goals supported
Progress persistence	Conversation lost	No	Full stateful dashboard
Built-in quizzes	No	No	Yes - per module, adaptive style
Community & sharing	No	No	Yes - connections, groups, sharing

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4. Technology Stack

Category	Technologies
Frontend	React 18, TypeScript, Vite, Tailwind CSS, shadcn/ui, TanStack React Query, React Router v6
Backend / Agents	Supabase Edge Functions (5 agents), Supabase Authentication, Supabase Postgres with RLS
AI / LLM	Google Gemini Pro (gemini-3-pro-preview), Gemini Flash (gemini-2.5-flash), Gemini Flash-Lite (gemini-2.5-flash-lite), Lovable AI Gateway

Category	Technologies
Search & Data APIs	Serper API (scrapes major search engines), YouTube Data API v3
Dev & Deployment	Lovable (React/Vite scaffold & deployment), Codex, Claude Code, GitHub

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5. System Architecture

WayVion is a full-stack web application: React 18 frontend, Supabase backend (Postgres, authentication, serverless Edge Functions), and external AI/search APIs. The architecture is organized around five distinct agent Edge Functions, each scoped to a single concern.

Architecture Layers

- The Surface (Client): React 18 + Vite frontend handling all user interaction, routing, and state.
- The Gatekeeper: Supabase Auth managing email/password, anonymous guest access, and password reset flows.
- The Intelligence: Five Supabase Edge Functions (AI agents) running Gemini models. Operates as a two-phase generative engine - Phase 1 structures the curriculum, Phase 2 curates resources in the background.
- The Memory: Supabase Postgres with five tables and Row Level Security, persisting roadmaps, progress, adaptations, profiles, and topic images.

Key Pages (Frontend)

Page / Route	Purpose
Landing (/)	Hero, How It Works, Why WayVion, Explore Categories, Footer with QR code
New Roadmap (/home, /new)	Topic input, skill level, timeline config, generation animation
Dashboard (/dashboard/:roadmapId)	Module list, progress tracking, quizzes, adaptation controls
My Roadmaps (/my-roadmaps)	All saved roadmaps including group-assigned ones
Flashcards (/flashcards)	Study tool generated from quiz content
Community (/community)	Topic-based learner discovery, connection management
Groups (/groups)	Classroom, team, and study group management
Profile (/profile)	Account settings, role toggle, guest-to-permanent conversion

Authentication

Three authentication flows via Supabase Auth:

- Email/Password: Standard signup with live validation (8+ chars, number, special character). Email confirmation required.
- Guest/Anonymous: signInAnonymously() for instant access. Can later convert to a permanent account in the Profile page.
- Forgot Password: resetPasswordForEmail() sends a link; user lands on /reset-password to set a new password.

A database trigger (handle_new_user) auto-creates a profiles row on signup. Guest users are filtered from community discovery.

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6. Four AI Agents in Five Edge Functions

WayVion is built on a multi-agent AI architecture where each agent has a distinct, scoped responsibility. Agents communicate through structured JSON, with Supabase as the persistence and orchestration layer.

#	Agent Name	Responsibility	Model Used
1	generate-roadmap	Curriculum architect - domain analysis, module structure, learning objectives, search query generation. Uses Gemini Pro directly (not via gateway) for maximum control.	Gemini Pro (gemini-3-pro-preview)
	adapt-roadmap	Quiz generator - creates 3–5 MCQs per module on demand, adapting question style to learning goal (practical vs. conceptual). Also powers flashcard generation.	
2	populate-resources	Resource curator - runs the full 10-stage curation pipeline integrating Serper and YouTube APIs. Triggered by the frontend after Phase 1 completes.	Gemini Flash (gemini-2.5-flash)
3	check-in	Performance analyzer - evaluates quiz score + self-report + time spent. Returns adaptation decision (none/minor/major) and updated roadmap if needed.	Gemini Flash (gemini-2.5-flash)
4	generate-module-quiz	Quiz generator - creates 3–5 MCQs per module on demand, adapting question style to learning goal (practical vs. conceptual). Also powers flashcard generation.	Gemini Flash-Lite (gemini-2.5-flash-lite)

AI Model Selection Rationale

All five agents initially defaulted to Gemini 2.5 Flash. After multiple iterations, model assignments were refined:

- Gemini Pro assigned to roadmap generation and both adaptation agents - curriculum design, prerequisite mapping, learning objective crafting, and structural adaptation are pedagogical reasoning tasks where Pro produces meaningfully better output.
- Gemini Flash assigned to resource curation and check-in - targeted scoring (~18 candidates/module) and performance evaluation do not require Pro-level reasoning; Flash delivers quality at lower latency.
- Gemini Flash-Lite assigned to quiz generation - 3–5 MCQs per module require minimal context; Flash-Lite is cost-effective and sufficient.

A switch to Gemini Flash for roadmap generation was considered to reduce latency but was decided against - latency impact is instead mitigated by the two-phase pipeline, which returns structure quickly regardless of model choice.

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7. Two-Phase Generation Pipeline

End-to-end roadmap generation initially took 20–25 seconds (and up to 1–1.25 minutes in earlier versions) with no user feedback. A two-phase delivery model was implemented and unified across all generation functions (generate-roadmap, adapt-roadmap, and check-in).

Phase	Description	Typical Latency
Phase 1 - Structure	Agent 1 (Gemini Pro) generates module structure only → saved to DB → immediately returned to user. Modules display with loading skeletons.	~8–30 seconds
Phase 2 - Resources	populate-resources runs the full Agent 2 + 10-stage pipeline in the background. Resources fill in automatically as they complete.	~10–15 seconds (background)

Previously, generate-roadmap, adapt-roadmap, and check-in each had separate, near-duplicate resource logic. All three now call the same populate-resources function, ensuring consistent resource quality everywhere and a single pipeline to maintain.

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8. The 10-Stage Resource Curation Pipeline

The most technically complex component of WayVion, implemented in a shared Supabase Edge Function (supabase/functions/_shared/resource-pipeline.ts). The pipeline runs 10 sequential and parallel stages - from fetching raw candidates via Serper and YouTube APIs, through hard filtering, authority scoring, parallel LLM pedagogical evaluation, cross-module negotiation, diversity clustering, and final reranking, to time-budget-constrained final assembly.

Pipeline Phases

The pipeline is organized into three phases:

- Phase A - Discovery (Stages 1–3): Context building, high-recall retrieval via Serper, YouTube metadata enrichment.
- Phase B - Critical Analysis (Stages 4–6): Hard filtering, authority scoring, AI context fit scoring.
- Phase C - Synthesis (Stages 7–10): Negotiation pass, clustering & diversity, batch LLM reranking, final assembly.

Stage-by-Stage Detail

Stage	Name	What Happens
1	Module Context	Builds a rich context object per module from user inputs: topic, skill level, learning objectives, prerequisites, and time budget. This context is passed to all subsequent stages.
2	High-Recall Retrieval	Serper API casts a wide net - 4–6 web results per module-specific search query. Query strategies differ by learning goal: more video queries for hands-on goals, more documentation queries for deep mastery.
3	YouTube Enrichment	YouTube Data API v3 fetches duration, view count, and channel metadata for all video results from Stage 2, enabling accurate per-resource time budgeting. Batch processed (50 IDs/call) to stay within daily quota.
4	Hard Filtering	Three sub-filters: (1) Anchor enforcement - topic terms must appear in title/description; (2) Scope Mismatch Penalty - rejects content too broad or too narrow; (3) Semantic threshold - resources must score ≥0.20 hybrid similarity to module title.
5	Authority Scoring	Tier-based trust via a Light Prior system (max +5 pts): Official Docs (1.00) > Vendor Docs (0.90) > University Papers (0.85) > Trusted YouTube (0.80) > Education Platform (0.75) > Blog (0.60) > Community/Reddit (0.42). Caps: max 40% video per module (60% for hands-on), max 40% documentation.
6	AI Context Fit	Gemini Flash evaluates each candidate on 5 pedagogical dimensions: semantic relevance, level alignment, goal alignment, pedagogical quality, tool/framework fit. One Flash call per module, all in parallel via Promise.all - wall time equals the slowest single module (~3–5s).
7	Negotiation Pass	Runs parallel with Stage 6. If two modules claim the same resource, the one with higher semantic similarity wins, ensuring no resource appears in more than one module.
8	Clustering & Diversity	Groups semantically similar resources; removes near-duplicates (>0.92 cosine similarity); enforces content-type variety per module.
9	Batch LLM Reranking	Final AI-powered ranking of shortlisted candidates per module, considering the full curriculum context (all module titles and objectives) to optimize sequencing.

Stage	Name	What Happens
10	Final Assembly	Assigns resources within time budget (85% usable time, 1.1x cap per module). Handles spanning resources - long videos across consecutive modules via span_plan. Validates continuation flags so 'Continue watching' only appears if the resource appeared in a prior module.

Serper Fetch Counts by Learning Goal

Learning Goal	Video Fetch / Web Fetch	Mix Target	Query Strategy
Conceptual	4 / 8	~40% video, ~60% articles	Adds 'explained with examples' queries
Hands-On	8 / 5	~60% video, ~40% articles	Tutorials, build, code-along queries
Quick Overview	5 / 6	~50/50 flexible	Adds 'cheat sheet', 'top things to know'
Deep Mastery	3 / 10	~25% video, ~75% articles	Adds arXiv, IEEE, MIT, Stanford queries

Parallel Per-Module Scoring

Rather than a single monolithic AI call with all candidates, one Agent 2 LLM call is launched per module, all in parallel via Promise.all. Key benefits:

- Wall time = slowest single module (~3–5s), not the sum of all modules (~15–30s for a large roadmap).
- Failure isolation - if one module's call times out, only that module falls back to heuristic scoring while all others complete with AI-curated resources (~90% of modules get high-quality AI-curated resources even under load).
- Focused context per call - each call handles ~18 candidates for one module, reducing cross-contamination risk.

Resource Coverage Backfill (85% Rule)

After the main assembly loop, each module's coverage is checked. If total resource minutes < 85% of the module's time budget, a backfill round runs:

- New Serper searches fire in parallel for under-filled modules using broader, objective-based queries (different from round 1 to avoid duplicate results).
- New results are YouTube-enriched, filtered, scored, and added until 85% coverage or the maxResources cap is reached.
- The 1–5 resource count rule is preserved - backfill is about time coverage, not count.

9. Database Schema

Five Supabase Postgres tables power WayVion's persistence layer. All tables enforce Row Level Security (RLS) policies restricting every query to the authenticated user's own data.

Core Tables

Table	Purpose
profiles	Stores user identity: display_name, bio, location, website, phone. Auto-created on signup via a database trigger (handle_new_user). Links to Supabase Auth user ID.
roadmaps	Core curriculum storage. roadmap_data (JSONB) holds complete module structure: module IDs, titles, descriptions, estimated_hours, day ranges, prerequisites, learning_objectives, and resources. Tracks status (active/completed/archived), streak count, completion counts, deadline. original_roadmap_data (JSONB) preserves the exact pre-adaptation version for rollback.
progress	Per-module tracking: roadmap_id (FK), module_id, status (not_started → in_progress → completed/skipped), quiz_score, self_report (easy/medium/hard), time_spent (minutes), completed_resources (array of resource URLs), notes (free text). Updated after each module interaction.
adaptations	Audit log of every roadmap change: roadmap_id (FK), previous_roadmap_data (JSONB), new_roadmap_data (JSONB), trigger_reason (check_in / manual_adapt), change_summary (text), created_at. Enables full rollback and change history.
topic_images	Caches AI-generated topic illustrations for the landing page and roadmap cards: topic_name (unique text key), image_url (text), created_at. Prevents regenerating images for recurring topics.

Community & Groups Tables (Post-Iteration)

Table	Purpose
connections	Replaces 'follows'. Fields: requester_id, receiver_id, status (pending | accepted | rejected). One row per pair; accepted = mutual connection.
groups	Group identity: name, type (classroom/team/study_group), owner_id, invite_code, is_active.
group_members	Membership: group_id, user_id, role (member | admin).
group_roadmaps	Roadmaps added to a group (not yet shared to members).
member_group_roadmaps	Junction table linking each member's cloned roadmap to the group assignment.

10. Core Feature Areas

10.1 Roadmap Generation & Resource Curation

Agent 1 (Gemini Pro) produces the curriculum structure. For a roadmap with 6 modules, 6 parallel Flash agents run simultaneously - one per module - for resource curation. Input parameters:

- Topic (free text)
- Skill level: Beginner / Intermediate / Advanced
- Learning goal: Quick Overview / Hands-On / Conceptual / Deep Mastery
- Timeline: hours/day × days/weeks

Module count is capped based on total hours to prevent over-fragmentation:

Total Hours	Max Modules	Min Hours Per Module
≤ 12h	floor(hours / 2)	2h
12–50h	floor(hours / 3)	3h
> 50h	floor(hours / 4)	4h

10.2 Adaptation - Check-In (Automatic)

After completing a module and taking a quiz, the check-in agent evaluates three signals:

- Self-report: easy / medium / hard (user-reported difficulty)
- Quiz score (system-generated)
- Time spent vs. estimated (passively tracked - users never manually input time)

Returns an AdaptationResult: none / minor / major, with reason and optionally an updated roadmap. Minor adaptations may add review modules or adjust upcoming module scope. Major adaptations may restructure remaining modules significantly. Completed modules are never modified.

10.3 Adaptation - Adapt Roadmap (Manual)

Users can manually request plan restructuring. The adapt modal exposes four controls:

- Remaining timeline (days/hours): The original control - recalculates the schedule around updated constraints.
  - Topic: Pre-filled with current topic. User can keep it, append keywords, or rewrite entirely. AI pivots remaining modules toward the new topic while preserving completed work.
- Learning Goal: 4-option grid (Conceptual / Hands-On / Quick Overview / Deep Mastery). Changing this adjusts resource type mix for remaining modules.

- Proficiency Level: 3-way toggle (Beginner / Intermediate / Advanced). Changing this adjusts depth and complexity of regenerated modules.

No new edge functions were needed for the topic/goal/proficiency controls - the existing adapt-roadmap function accepts two new optional parameters (new_topic, new_skill_level) and the AI prompt was updated accordingly. Users receive multiple restructuring options and can roll back to the previous roadmap if unsatisfied.

10.4 Quizzes & Flashcards

A dedicated agent (Agent 5, Gemini Flash-Lite) generates 3–5 multiple-choice questions per module on demand, adapting question style to the learning goal (practical for hands-on, conceptual for theory). Quiz style:

- Practical/hands-on goal → code-based and application questions
- Conceptual/deep mastery goal → explanation and reasoning questions

Flashcards are generated from quiz content. On first visit to the Flashcards page, quizzes are auto-generated for all modules missing them (batched 3 at a time, ~15–30s). Results are persisted to the database so subsequent visits are instant.

10.5 Community of Learners

A full social layer allows learners to discover each other, connect, and share roadmaps.

Connection Model

LinkedIn-style mutual connections (replacing an earlier Twitter-style one-way following design). When one user accepts a connection request, both parties are connected - no second request needed. UI states: Connect → Pending → Connected.

Points System

Points are computed on-the-fly via a Postgres function (no trigger needed):

- +10 points per completed module
- +25 points per created roadmap
- +5 points per roadmap shared and accepted by a recipient

Discovery & Profiles

- Public profiles at /user/:userId show roadmaps (topic, skill level, timeline) but never progress data.
- /community page allows topic-based search - search 'Machine Learning' to find other learners on the same topic.

- Default visibility: profiles are public by default (matching LinkedIn convention); guest accounts are filtered from discovery.
- My Connections section shows accepted connections; Find Other Learners shows all other public profiles.

Roadmap Sharing Flow

- Share button lives inside the roadmap dashboard alongside Adapt and Revert.
  - Sender selects one or more connections via a multi-select modal.
  - Receiver sees the full roadmap preview (all modules) before deciding.
    - Accept: roadmap is cloned fresh to receiver with zero progress.
    - Reject: request disappears; sender's roadmap is untouched.
      - Sender earns +5 points only when the receiver accepts.

Roadmap Request Flow

From another user's public profile, you can request a specific roadmap they own. The owner can accept (share it) or reject. This is the reverse of the share flow - pull instead of push.

10.6 Groups - Team-Based & Classroom Learning

A generic groups system supports structured, multi-user learning in three contexts:

Group Type	Context & Labels
classroom	Educator creates; members = Students; owner = Professor
team	Manager creates; members = Members; owner = Manager
study_group	Any learner creates; all terms = generic Admin/Member

Profile Roles

Users have a role field (learner / educator / manager) toggleable in Profile Settings. All roles can create study groups. Educator and Manager roles unlock classroom and team creation plus the progress dashboard.

Joining a Group

- Invite code (e.g. XKFM-29NP) - for in-person distribution
  - Invite link (/join/XKFM-29NP) - for online sharing
- Late joiners automatically receive all previously shared roadmaps

Roadmap Assignment Flow

- Owner adds up to 5 roadmaps to a group (via dropdown or from within any roadmap's Share modal).
- Adding a roadmap does NOT auto-assign it - owner must explicitly click Share.
- Share clones the roadmap to all current members instantly; Share button becomes disabled and shows 'Shared'.
- Members access assigned roadmaps from a 'Group Roadmaps' section in My Roadmaps.
- Each member's copy is fully independent: separate progress, adaptation, quiz scores.

Progress Dashboard (Owner View)

The group owner can open a progress dashboard per assigned roadmap showing:

- Summary cards: total members, average completion %, average quiz score, count of members reporting a module as hard.
- Member table (sortable): completion %, quiz average, time spent, hard modules, last active.
- Expandable per-module breakdown per member: status, quiz score, time, difficulty, completion timestamp.
- Color-coded completion bar chart across all members.
- Module Difficulty Insights: which modules members found hard vs. easy.

Edge Cases Handled

- Owner deletes account → CASCADE deletes group and memberships.
- Owner deletes original roadmap → source_roadmap_id becomes NULL; member copies continue unaffected.
- Guest users cannot create or join groups.
- Owner can regenerate or deactivate invite code if leaked.

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11. Key Technical Iterations

WayVion underwent five major technical iterations during the MSIS 549B period, plus additional architectural changes post-milestone.

Iteration	Change	Before	After	Impact
1	Rule-Based to LLM-Powered Resource Scoring	Heuristic scoring: keyword matching, view count, domain authority. Led to 80–90% video dominance and off-topic resources in wrong modules.	Hybrid: heuristics filter candidates, LLM evaluates pedagogical fit on 5 dimensions.	Off-topic contamination eliminated; content-type distribution normalized.

Iteration	Change	Before	After	Impact
2	Gemini API Keys vs. Lovable Gateway	All LLM calls routed through Lovable Gateway, adding significant latency.	Direct Gemini API keys for Agents 1 and 2; gateway as fallback.	Saved ~20 seconds on generation time.
3	Monolithic to Parallel Per-Module Resource Scoring	Single large AI call with all modules and all candidates (~15–30s, cross-contamination risk, catastrophic failure mode).	One Flash call per module, all in parallel via Promise.all.	Wall time reduced to ~3–5s; failure isolation achieved.
4	Token Reduction for Adaptation Prompts	Full resource/quiz arrays sent for completed modules (~10 fields × 3–5 resources per module).	Stripping resources/quiz arrays from completed modules sent to AI (frontend merges originals back after response).	~20–30% token reduction; 1–3 seconds faster per adaptation call.
5	Sequential to Two-Phase Generation	Single-pass generation (20–25s, no user feedback). Prior attempt at splitting failed silently due to fire-and-forget.	Phase 1 returns module structure (8–15s); Phase 2 populates resources in background via explicit frontend-driven call with fetchData() on completion.	Perceived performance significantly improved; silent failure mode resolved.

Post-Milestone Changes

Change	Impact Level	Description
Community of Learners	High	Full social layer: connections, public profiles, topic-based discovery, roadmap sharing and requesting.
Two-phase pipeline unification	High	All three generation functions (generate-roadmap, adapt-roadmap, check-in) now call the same populate-resources function.
Groups system	High	Classroom, team, and study group contexts with owner progress dashboards and independent member copies.
Adapt roadmap - extended controls	High	Topic, learning goal, and proficiency level added to the adapt modal (~80% UI effort, ~20% prompt adjustments).
Module removal	Medium	Users can remove individual modules directly from the dashboard.

Change	Impact Level	Description
Resource coverage backfill (85% rule)	Medium	Backfill round fires if total resource minutes < 85% of module time budget.
Resource blocking - roadmap-scoped	Low	Changed from topic-scoped (too aggressive) to roadmap-scoped blocking.
Model selection review	Low	Confirmed Gemini Pro for roadmap generation; latency mitigated by two-phase pipeline.

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12. Key Trade-Off Decisions

Decision	Chosen Approach	Alternative Considered	Trade-off Accepted
Roadmap Generation model	Gemini Pro	Gemini 2.5 Flash	~30s added latency; accepted for superior curriculum quality as the foundation everything else builds on.
Resource Curation model	Gemini Flash	Gemini Pro	Slightly lower reasoning depth; accepted because resource scoring is a targeted, well-scoped task.
Quiz Generation model	Gemini Flash-Lite	Gemini Flash	Minimal context needed for 3–5 MCQs; Flash-Lite is cost-effective and sufficient.
Check-in model	Gemini Pro	Gemini Flash	Check-in can trigger structural changes requiring deeper reasoning; Pro kept despite latency.
Resource diversity cap	Max 60% video per module (40% for non-hands-on)	No cap	May exclude a good video; guarantees content-type mix.
Module count limit	Capped by floor(hours/N)	No limit	Less granularity; prevents AI from creating 15 tiny 30-minute modules.
Serper cache TTL	48 hours	No cache	May miss newly published content; saves significant API costs.
YouTube metadata cache TTL	168 hours (1 week)	Shorter TTL	View counts may be stale; YouTube quota is expensive.
Connection model	LinkedIn-style mutual	Twitter-style one-way following	Slightly higher friction to connect; produces higher-quality social graph for learning communities.

13. Performance Summary

Operation	Typical Latency	Bottleneck
Generate Roadmap (full, end-to-end)	35–65 seconds	Agent 1 (Gemini Pro) + resource pipeline
Adapt Roadmap (skip_resources)	8–15 seconds	Agent 1 (Gemini Pro) AI call
Resource Population (background)	10–15 seconds	Serper + YouTube + Agent 2
Quiz Generation (per module)	3–5 seconds	Gemini Flash-Lite model
Flashcards (first visit, all modules)	15–30 seconds	Batch quiz generation
Flashcards (subsequent visits)	Instant	Cached in DB
Per-module resource scoring (parallel)	~3–5 seconds	Slowest single module Flash call

14. Results & Evaluation

Three test cases were run against a 5-dimension scoring rubric (each scored 1–5; 5 = target fully met).

Scoring Rubric

Metric	5 - Excellent	4 - Good	Target
Roadmap Coherence	Logical prereq sequence; all topics relevant and well-ordered	Minor gaps or minor ordering issues	Score ≥ 4
Resource Relevance	≥ 80% of resources on topic for their assigned module	60–79% on-topic	≥ 80% relevant
Resource Quality	≥ 70% from high-authority tier (score ≥ 0.75)	50–69% high-authority	≥ 60% high-authority
Content-Type Diversity	No single type exceeds 60% per module	One type is 60–75%	No single type > 60%
Generation Reliability	100% end-to-end success, no errors	1–2 minor recoverable errors	≥ 95% success

Test Case Results

Test Case	Input	Score	Notes
TC1 - Standard	Finance & Investment | Intermediate | Conceptual | 2 weeks, 1h/day (14h)	23.5 / 25	Strong coherence and content-type diversity; resource quality varied slightly across modules.
TC2 - Very Short	Open Claw | Beginner | Quick Overview | 2 hours	24 / 25	Resources precisely matched topic and fit within 2 hours; near-perfect across all dimensions.
TC3 - Niche Edge Case	Quantifying Behavioral Drift in LLMs | Advanced | Deep Research | 6 weeks, 4h/day (168h)	18.5 / 25	Primary failure mode: pipeline found resources covering only ~71 of 168 hours required. Sparse web content on this topic forced lower-tier sources in some modules.

Failure Analysis: The primary failure mode revealed by Test Case 3 is resource coverage for niche topics with long timelines. The topic 'Quantifying Behavioral Drift in LLMs' has limited indexed content on the internet. The agent assigns whatever relevant resources it can find but cannot fabricate coverage that does not exist on the web - a deliberate design decision to avoid hallucination, though it leaves visible gaps for rare topics.

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15. Known Limitations

- Generation Latency: End-to-end roadmap generation takes 35–65 seconds. Two-phase delivery shows modules in 8–15s, but full resource population is slower due to Gemini Pro latency and the multi-stage pipeline.
- API Cost at Scale: A 10-module roadmap triggers ~10–15 external API calls (Gemini Pro + Flash, Serper, YouTube). At scale, cost management would require job queuing, aggressive caching, and a dedicated compute layer.
- No Real-Time Resource Validation: Links are verified at retrieval time but not monitored for continued availability. A production system requires periodic link health checks.
- Sparse Coverage for Niche Topics: Highly specialized topics with limited web content produce roadmaps with resource gaps, as demonstrated in Test Case 3.
- Adaptation Signal Quality: Check-in agents rely on user self-reports and quiz scores, which are imperfect proxies. Deeper behavioral signals (time-on-resource, skip patterns) would improve adaptation accuracy.
- No Real Content Visibility: WayVion fetches metadata from Serper and enriches via YouTube API but has no visibility into actual content quality - only metadata signals like titles, descriptions, view counts, and channel authority.

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16. Future Work

- Live Collaboration & Sharing: Allow teams to share roadmaps with permission controls and collaborative progress tracking (foundation now exists via Groups system).
- Learning Topic Extension: Allow users to add related topics to an existing roadmap mid-way (e.g., adding Machine Learning to an existing AI roadmap).
- Resource Feedback Loop: Let users rate resources; feed ratings back into the scoring model for continuous improvement.
- Streaming Generation: Surface module titles one-by-one as the LLM generates them, reducing perceived latency further.
- Mobile App: Native mobile experience with offline access and push-based streak reminders.
- Content Quality Signals: Approaches under consideration include transcript-based scoring, article content fetching, user feedback loops, and completion rate tracking as a behavioral proxy.
- 'Why This Resource?' Explainer: Surface the LLM scoring rationale already present in the pipeline on individual resource cards - identified as a high-impact, demo-ready feature improvement.

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17. Ethical Considerations

Source Bias & Information Quality

WayVion's authority tier favors established institutions (MIT, Stanford, official documentation) and verified YouTube channels. While this reduces misinformation risk, it may systematically under-rank accurate content from lesser-known creators or non-English sources. A production system should audit for geographic and language bias in recommendations.

Hallucination Risk

Early versions allowed LLMs to suggest resource URLs, which led to hallucinated links. The current architecture eliminates this by separating URL discovery (Serper/YouTube return real URLs) from LLM evaluation (which scores only, never invents URLs). This separation is a deliberate ethical and reliability safeguard.

AI Dependency & Learner Agency

The adaptation feature lowers friction for falling behind, but risks removing productive struggle - the cognitive effort that drives deep learning. The system mitigates this by surfacing adaptation as options with explicit tradeoffs, not silent restructuring. Users retain full agency over whether to accept changes or roll back.

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18. References & Code

GitHub Repository: https://github.com/swatt145-png/pathfinder-ai-roadmap

AI Disclosure: During report writing, Claude (Anthropic) was used to assist with report structure, checking and rephrasing content to fit page length. All technical content, architectural decisions, and implementation details are the author's own.

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